

Assessment 2: Individual Lab Report

Introduction:

Data on carbon dioxide (CO₂) emissions per capita from 2005 to 2024 have been compiled by the World Bank for countries worldwide. In preparation for an international Climate Change Conference, this report aims to analyse historical patterns in CO₂ emissions, with a particular focus on the United Kingdom. Understanding emission trends is essential for evaluating the effectiveness of environmental policies and assessing progress toward climate targets.

The primary objective of this analysis is to investigate how UK CO₂ emissions have changed over time and to determine whether recent emissions differ significantly from previous years. In addition, comparisons will be made between the UK and the European Union average, as well as between countries of different income levels, to assess whether broader economic factors are associated with changes in emissions.

Given the urgency of global climate challenges, even moderate differences in emissions can have substantial long-term environmental impacts. Therefore, statistical analysis is used to determine whether observed differences are likely due to genuine underlying changes rather than random variation. This report will combine exploratory analysis with formal hypothesis testing to provide evidence-based conclusions regarding trends in CO₂ emissions.

Exploratory Analysis:

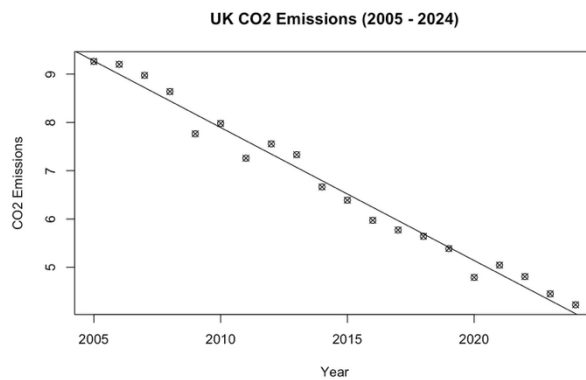


Figure 1: Timeline of the CO₂ Emissions in the UK.

Figure 1 shows a clear downward trend over the period in UK CO₂ emissions. Emissions have fallen from 9.263 metric tonnes per capita in 2005 to 4.224 metric tonnes per capita by 2024. Using R a regression line has been plotted with gradient -0.2755, which suggests that there is approximately a fall of 0.2755 metric tonnes per capita of CO₂ emissions for every year that passes.

Minimum	1 st Quartile	Median	3 rd Quartile	Maximum
4.224	5.302	6.527	7.817	9.263

Figure 2: Five number summary for CO₂ emissions in the UK.

Figure 2 is a five number summary from which we can find an IQR of 2.515 metric tonnes per capita, which indicates moderate spread over the years. The median lies close to the lower quartile than the upper one, this could reflect the overall downward trend.

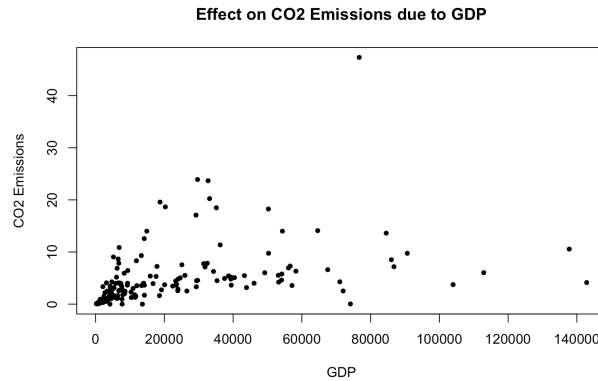


Figure 3: Scatterplot of CO₂ emissions against the GDP of countries from 2024.

Figure 3 shows that countries with a lower GDP tend to cluster at lower levels of CO₂ emissions, however as the GDP increases the CO₂ emissions show much greater variability. Thus, the scatterplot shows no obvious linear trend. Also there seem to be a few outliers for example the country with a moderately high GDP but extremely high CO₂ emissions, as well as countries with very high GDPs but moderate levels of CO₂ emissions.

Statistical Analysis:

Comparison of Variances Between Decades:

The hypotheses that are to be tested are:

$$H_0 : \sigma^2_{\text{Old}} = \sigma^2_{\text{New}}$$

$$H_1 : \sigma^2_{\text{Old}} \neq \sigma^2_{\text{New}}$$

To test whether the variability in UK CO₂ emissions differ between 2005 – 2014 and 2015 – 2024, a test for the equality of variances when sampling from two independent normal distributions will require an F – test.

After computation in R, we find that the p – value from this test to be 0.4525, which is substantially greater than a significance level of 0.05. Therefore, there is insufficient evidence to reject the null hypothesis. This suggests that there is a statistically insignificant difference in the variance of UK CO₂ emissions between the two decades.

Comparison of Means Between Decades:

The hypotheses that are to be tested are:

$$H_0 : \mu_{\text{Old}} - \mu_{\text{New}} = 0$$

$$H_1 : \mu_{\text{Old}} - \mu_{\text{New}} > 0$$

To assess whether the mean CO₂ emissions in 2005 – 2014 are higher than in 2015 – 2024, we will apply a two-sample pooled t-test, since samples are from two independent normal distributions and previous analysis allows us to assume equal variances.

After performing the t-test in R we find that the p value is 1.836×10^{-7} which is extremely small and below the 0.05 significance level. Therefore, there is strong significant statistical evidence to say that the mean UK CO₂ emissions in the previous decade are lower than the preceding decade.

Comparison of Means Between the UK and the EU:

The hypotheses that are to be tested are:

$$H_0 : \mu_{\text{UK}} - \mu_{\text{EU}} = 0$$

$$H_1 : \mu_{\text{UK}} - \mu_{\text{EU}} < 0$$

To find whether the mean CO₂ emissions of the UK are lower than the average emissions of the EU, we will perform a Welch test. This is due to the variances being assumed unequal as well them being normally distributed.

After calculation using R, we find the p value to be 0.1754, this is greater than the 0.05 significance level. Therefore, there is insufficient statistical evidence to conclude that UK CO₂ emissions are lower than the EU average over the period, even though the UK average is slightly lower at 6.655 compared with the EU average of 7.053.

Relationship Between Income Level and Change in CO₂ Emissions:

The hypotheses to be tested are:

H_0 : Income level and change in CO₂ emissions are independent

H_1 : Income level and change in CO₂ emissions are not independent

To assess whether the change in average CO₂ emissions is associated with income level of a country, we will use a Chi-squared test for independence using the data in Table 1.

After computation in R, we have received a p value of 1.049×10^{-8} which is extremely small, and far below the 0.05 significance level. As a result, there is significant statistical evidence to say that income level is associated with the income level of a country.

Conclusion:

Both exploratory and formal statistical analysis indicate a clear reduction in UK CO₂ emissions between 2005 and 2024. The timeline plot showed a strong downward trend, with emissions falling from 9.263 to 4.224 metric tonnes per capita and an estimated annual decrease of approximately 0.276 metric tonnes per capita. This visual pattern was supported by the two-sample t-test, which provided strong statistical evidence that mean emissions in 2015–2024 are significantly lower than in 2005–2014. The F-test confirmed that variability between decades did not differ significantly.

Although the UK's average emissions were slightly lower than the European Union average, the Welch test showed this difference was not statistically significant. In contrast, the chi-squared test demonstrated a strong association between income level and changes in emissions with high-income countries more likely to experience reductions.

Overall, the results confirm that the UK has achieved a statistically significant decline in CO₂ emissions over the past two decades, while broader global patterns appear to vary according to economic classification.